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RESEARCH AREA

- **Efficient AI Systems**
Exploring efficient machine learning for edge devices, with an emphasis on hardware-aware optimization and FPGA-based acceleration.
- **Computer Vision**
Developing computer vision algorithms for visual perception, focusing on 2D and 3D object detection from images and depth sensors.
- **Intelligent Sensing**
Integrating sensors, signal processing, and data analytics to extract meaningful information from physical environments.

PROFESSIONAL APPOINTMENTS

2024.01 – present	University Lecturer <i>Politeknik Elektronika Negeri Surabaya, Surabaya, Indonesia</i>
2024.02 – 2024.07	Senior Hardware & Internet of Things (IoT) Engineer <i>McEasy - PT. Otto Menara Globalindo, Surabaya & Jakarta, Indonesia</i>
2023.05 – 2024.01	Hardware & Internet of Things (IoT) Engineer <i>McEasy - PT. Otto Menara Globalindo, Surabaya & Jakarta, Indonesia</i>
2023.05 – 2024.01	Hardware & Internet of Things (IoT) Engineer <i>McEasy - PT. Otto Menara Globalindo, Surabaya & Jakarta, Indonesia</i>
2023.02 – 2023.06	Research Assistant - Collision Avoidance System Developmnet For Mining <i>School of Electrical and Informatics Engineering - Institut Teknologi Bandung, Bandung, Indonesia</i>
2023.02 – 2023.06	Research Assistant - Collision Avoidance System Developmnet For Mining <i>School of Electrical and Informatics Engineering - Institut Teknologi Bandung, Bandung, Indonesia</i>
2018.03 – 2020.08	Internet of Things (IoT) Engineer <i>McEasy - PT. Otto Menara Globalindo, Surabaya, Indonesia</i>
2017.04 – 2018.02	Internet of Things (IoT) Engineer (Part-time) <i>McEasy - PT. Otto Menara Globalindo, Surabaya, Indonesia</i>
2015.08	Internship <i>PT. Pertamina Hulu Energi West Madura Offshore, Gresik, Indonesia</i>

EDUCATION

M.Sc.	Control and Intelligent Systems	<i>Institut Teknologi Bandung, Indonesia</i>	2020 – 2023
B.ASc.	Electronics Engineering	<i>Politeknik Elektronika Negeri Surabaya, Indonesia</i>	2017 – 2018
A.Md.	Electronics Engineering	<i>Politeknik Elektronika Negeri Surabaya, Indonesia</i>	2013 – 2016

RESEARCH FUNDING

2026 – Present

Test

Role: PI

2025 – Present.

Robot AGV (Automated Guided Vehicle)

Role: Co-PI, **PI:** Prof. Dr. Ir. Dedid Cahya Happyanto, M.T.

2025 – Present.

Pengembangan Mobil Listrik Otonom Berbasis Internet of Things dengan Sistem Pengendalian Cerdas dan Monitoring Keselamatan Terintegrasi

Role: Co-PI, **PI:** Prof. Dr. Ir. Dedid Cahya Happyanto, M.T.

2025.

Implementasi Early Warning System berbasis MQTT untuk Deteksi Dini Kualitas Air Udang Vaname

Role: Co-PI, **PI:** Nirwana Haidar Hari, S.Pd., M.kom.

2025.

YOLOv11-Based Detection of Indonesian Traffic Signs: Transfer Learning vs. From-Scratch Training

Role: PI

2024 – 2025.

SMART-UHT: Otomatisasi Produksi Susu UHT menunjang Kemandirian Ekonomi Lokal di Puduk Ponorogo

Role: Co-PI, **PI:** Dr. Ir. Bambang Sumantri, S.T., M.Sc.

2023.

Collision Avoidance System Development For Mining

Role: Researcher, **PI:** Prof. Dr. Ir. Bambang Riyanto Trilaksono

2021 – 2023.

Autonomous Tram Development Using Artificial Intelligence

Role: Researcher, **PI:** Prof. Dr. Ir. Bambang Riyanto Trilaksono

PUBLICATIONS

Journals

- [J1] N. Tamami, **I. C. Ramadhan**, K. Alfathdyanto, Madyono, H. Hermawan, H. M. Rosalinda, A. W. R. Gusti, and W. Trisniani, “Implementasi Media Informasi Berbasis Running Text untuk Meningkatkan Efisiensi dan Efektivitas Penyampaian Informasi di Kelurahan Keputih,” *j-dinamika*, vol. 10, no. 2, pp. 254–260, Aug. 2025.
- [J2] **I. C. Ramadhan**, A. Hendriawan, and H. Oktavianto, “YOLOv11-Based Detection of Indonesian Traffic Signs: Transfer Learning vs. From-Scratch Training,” *Journal of Applied Informatics and Computing (JAIC)*, vol. 9, no. 4, pp. 1363–1373, Aug. 2025.

Conferences

- [C1] **I. C. Ramadhan**, B. R. Trilaksono, and E. M. I. Hidayat, “Towards Real-Time Graph Neural Network-Based 3D Object Detection for Autonomous Vehicles,” in *Proc. 19th IEEE Int. Colloq. Signal Processing & Its Applications (CSPA)*, Kedah, Malaysia, 2023, pp. 105–110.